

ASSIGNMENT NO.6: Data Manipulation and Table Management

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Questions:

Q1) Create a MySQL database named TY_CollegeDB and create two tables: Course and Student.

Course Table

Column Name	Data Type	Constraint
CourseID	INT	PRIMARY KEY
CourseName	VARCHAR(50)	NOT NULL
Fees	DECIMAL(10,2)	NOT NULL

Student Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
Age	INT	
City	VARCHAR(30)	
CourseID	INT	FOREIGN KEY

QUERY:

```

1 CREATE DATABASE TY_CollegeDB;
2
3 USE TY_CollegeDB;
4
5 CREATE TABLE Course (
6     CourseID INT PRIMARY KEY,
7     CourseName VARCHAR(50) NOT NULL,
8     Fees DECIMAL(10,2) NOT NULL
9 );
10
11 CREATE TABLE Student (
12     StudentID INT PRIMARY KEY,
13     StudentName VARCHAR(50) NOT NULL,
14     Age INT,
15     City VARCHAR(30),
16     CourseID INT,
17     FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
18 );
19 SHOW DATABASES;
20
21 USE TY_CollegeDB;
22
23 SHOW TABLES;
24
25 DESC Course;
26
27 DESC Student;

```

OUTPUT:

Output 1: Course Table Structure

Field	Type	Null	Key	Default	Extra
CourseID	int(11)	NO	PRI	NULL	
CourseName	varchar(50)	NO		NULL	
Fees	decimal(10,2)	NO		NULL	

Output 2: Student Table Structure

Field	Type	Null	Key	Default	Extra
StudentID	int(11)	NO	PRI	NULL	
StudentName	varchar(50)	NO		NULL	
Age	int(11)	YES		NULL	
City	varchar(30)	YES		NULL	
CourseID	int(11)	YES	MUL	NULL	

Q2) Insert at least 4 records into the Course table and 6 records into the Student table.
Make sure every student has a valid CourseID from the Course table.

QUERY:

```

1 USE TY_CollegeDB;
2
3 INSERT INTO Course (CourseID, CourseName, Fees)
4 VALUES
5 (101, 'Data Science', 55000.00),
6 (102, 'Computer Science', 50000.00),
7 (103, 'Information Technology', 48000.00),
8 (104, 'Artificial Intelligence', 60000.00);
9
10 INSERT INTO Student (StudentID, StudentName, Age, City, CourseID)
11 VALUES
12 (1, 'Aarav Sharma', 20, 'Pune', 101),
13 (2, 'Priya Patil', 21, 'Mumbai', 102),
14 (3, 'Rahul Joshi', 20, 'Pune', 101),
15 (4, 'Sneha Kulkarni', 22, 'Nashik', 103),
16 (5, 'Aditya Deshmukh', 21, 'Nagpur', 104),
17 (6, 'Neha Shah', 20, 'Pune', 102);
18
19 SELECT * FROM Course;
20 SELECT * FROM Student;

```

OUTPUT:

Output 1 Course table:

	CourseID	CourseName	Fees
☐ Edit ☐ Copy ☐ Delete	101	Data Science	55000.00
☐ Edit ☐ Copy ☐ Delete	102	Computer Science	50000.00
☐ Edit ☐ Copy ☐ Delete	103	Information Technology	48000.00
☐ Edit ☐ Copy ☐ Delete	104	Artificial Intelligence	60000.00

Output 2 Student table:

		StudentID	StudentName	Age	City	CourseID
☐	Edit Copy Delete	1	Aarav Sharma	20	Pune	101
☐	Edit Copy Delete	2	Priya Patil	21	Mumbai	102
☐	Edit Copy Delete	3	Rahul Joshi	20	Pune	101
☐	Edit Copy Delete	4	Sneha Kulkarni	22	Nashik	103
☐	Edit Copy Delete	5	Aditya Deshmukh	21	Nagpur	104
☐	Edit Copy Delete	6	Neha Shah	20	Pune	102

Q3) Write a MySQL query to insert one new student record into the Student table.

The record should include:

Field
StudentID
StudentName
Age
City
CourseID

QUERY:

```
1 INSERT INTO Student (StudentID, StudentName, Age, City, CourseID)
2 VALUES (7, 'Riya More', 21, 'Pune', 103);
3 SELECT * FROM Student;
```

OUTPUT:

		StudentID	StudentName	Age	City	CourseID
☐	Edit Copy Delete	1	Aarav Sharma	20	Pune	101
☐	Edit Copy Delete	2	Priya Patil	21	Mumbai	102
☐	Edit Copy Delete	3	Rahul Joshi	20	Pune	101
☐	Edit Copy Delete	4	Sneha Kulkarni	22	Nashik	103
☐	Edit Copy Delete	5	Aditya Deshmukh	21	Nagpur	104
☐	Edit Copy Delete	6	Neha Shah	20	Pune	102
☐	Edit Copy Delete	7	Riya More	21	Pune	103

Q4) Write a MySQL query to update the city of a student using StudentID.

Example requirement:

Change Required

Update the city of StudentID 2 to Mumbai

QUERY:

```
1 UPDATE Student
2 SET City = 'Mumbai'
3 WHERE StudentID = 2;
4
5 SELECT * FROM Student
6 WHERE StudentID = 2;
```

OUTPUT:

	StudentID	StudentName	Age	City	CourseID
☐ Edit <input type="plus"/> Copy <input type="minus"/> Delete	2	Priya Patil	21	Mumbai	102

Q5) Write a MySQL query to update the fees of a course using CourseID.

Example requirement:

Change Required

Update the fees of CourseID 101 to 47000

QUERY:

```
1 UPDATE Course
2 SET Fees = 47000.00
3 WHERE CourseID = 101;
4
5 SELECT * FROM Course
6 WHERE CourseID = 101;
```

OUTPUT:

	CourseID	CourseName	Fees
☐ Edit <input type="plus"/> Copy <input type="minus"/> Delete	101	Data Science	47000.00

Q6) Write a MySQL query to delete one student record using StudentID.

Also write one line explaining why the WHERE clause is important while using DELETE.

QUERY:

```
1 DELETE FROM Student
2 WHERE StudentID = 7;
3
4 SELECT * FROM Student;
```

OUTPUT:

						StudentID	StudentName	Age	City	CourseID	
☐		Edit		Copy		Delete	1	Aarav Sharma	20	Pune	101
☐		Edit		Copy		Delete	2	Priya Patil	21	Mumbai	102
☐		Edit		Copy		Delete	3	Rahul Joshi	20	Pune	101
☐		Edit		Copy		Delete	4	Sneha Kulkarni	22	Nashik	103
☐		Edit		Copy		Delete	5	Aditya Deshmukh	21	Nagpur	104
☐		Edit		Copy		Delete	6	Neha Shah	20	Pune	102

Explanation

The WHERE clause is important because it ensures that only the specified student record is deleted; without it, DELETE would remove all records from the table.